

Theorem 86 (Artin–Schreier–Witt theory; [Tho05, §3–4]). *Let $M \supseteq \mathbb{F}_p$ be a field. There is a canonical isomorphism*

$$W_r(M)/\wp W_r(M) \xrightarrow{\cong} H^1(M, \mathbb{Z}/p^r\mathbb{Z}).$$

The class of $\vec{g}$ corresponds to the torsor

$$\mathrm{Spec} M[\vec{X}]/(\wp \vec{X} \ominus \vec{g}).$$

If the class has order p^s , let $H \cong \mathbb{Z}/p^s\mathbb{Z}$ be the image of the associated character, equivalently the unique subgroup of order p^s in $\mathbb{Z}/p^r\mathbb{Z}$. The corresponding cyclic field extension L/M has degree p^s , and the torsor is the disjoint union of p^{r-s} connected components, each isomorphic to $\mathrm{Spec} L$ and each an H -torsor. Equivalently, the original $\mathbb{Z}/p^r\mathbb{Z}$ -torsor is induced from any one of these H -torsors. In particular, it is connected precisely when its class has order p^r . Every cyclic extension of degree dividing p^r arises in this way.

Proof. Let M^s be a separable closure of M and $\Gamma = \mathrm{Gal}(M^s/M)$. The triangular equations in the preceding proof show that $\wp$ is surjective on $W_r(M^s)$ and that its kernel is $W_r(\mathbb{F}_p)$. The connecting map sends the class of $\vec{g}$ to

$$\sigma \mapsto \sigma(\vec{u}) \ominus \vec{u}, \quad \wp \vec{u} = \vec{g}.$$

This is a continuous homomorphism $\Gamma \rightarrow W_r(\mathbb{F}_p)$, independent of the choices. It is injective on classes because a Γ -invariant solution lies in $W_r(M)$. It is surjective by additive Hilbert 90, applied inductively to

$$0 \longrightarrow M^s \xrightarrow{V^{r-1}} W_r(M^s) \xrightarrow{t} W_{r-1}(M^s) \longrightarrow 0.$$

Thus $W_r(M)/\wp W_r(M) \cong H^1(M, \mathbb{Z}/p^r\mathbb{Z})$. Write $H = \mathrm{im}(\chi)$ for the image of the associated character. Under the correspondence between finite constant-group torsors and continuous characters, the Galois orbits on the geometric fiber $\mathbb{Z}/p^r\mathbb{Z}$ are precisely the cosets of H . Hence there are $[\mathbb{Z}/p^r\mathbb{Z} : H] = p^{r-s}$ connected components; the stabilizer H acts simply transitively on each one, whose residue field is the degree- p^s extension cut out by $\ker(\chi)$. This gives the assertions about degree, connectedness, and induction from an H -torsor. $\square$

Corollary 87. *Let $M \supseteq \mathbb{F}_p$ and let $\vec{g} \in W_r(M)$ have $g_0 = 0$. Then*

$$p^{r-1}[\vec{g}] = 0 \quad \text{in } W_r(M)/\wp W_r(M).$$

Consequently, a class of order p^r has nonzero zeroth component in every representative.

Proof. By (A.8), $p^{r-1}\vec{g} = (0, \dots, 0, g_0^{p^{r-1}}) = 0$ already in $W_r(M)$. $\square$

A.4.3 Reduced polynomial representatives

Definition 88. A vector $\vec{g} = (g_0, \dots, g_{r-1}) \in W_r(k((x)))$ is a **reduced polynomial vector** if it is reduced and every $g_j \in x^{-1}k[x^{-1}]$.

Lemma 89 (Surjectivity on integral Witt vectors). *If k is algebraically closed, then*

$$\wp : W_r(k[[x]]) \longrightarrow W_r(k[[x]])$$

is surjective.